

Project 1

Title: Movie Rating Analysis Using Python

Objective:-

1. Analyze movie ratings using Python.
2. Calculate the average movie rating.
3. Identify the highest-rated and lowest-rated movies.
4. Display movies with ratings greater than 8.
5. Sort movies based on ratings.

Tools Used:-

Python 3

Google Colab or Jupyter Notebook

Pandas Library

Requirements:-

Python

Pandas

Install Pandas if needed:

pip install pandas

Procedure:-

Import the Pandas library.

Create a movie dataset.

Convert the dataset into a DataFrame.

Display the dataset.

Calculate the average rating.

Find the highest-rated movie.

Find the lowest-rated movie.

Display movies with ratings above 8.

Sort the movies by rating.

Save the results to a CSV file.

Program

```
import pandas as pd

movies = { "Movie": [ "Avatar", "Titanic", "Inception", "Interstellar", "The Dark Knight" ],
           "Rating": [7.8, 7.9, 8.8, 8.7, 9.0],
           "Year": [2009, 1997, 2010, 2014, 2008]
}

df = pd.DataFrame(movies)

print("Movie Dataset")

print(df)

print("\nAverage Rating")

print(df["Rating"].mean())

print("\nHighest Rated Movie")

print(df.loc[df["Rating"].idxmax()])

print("\nLowest Rated Movie")

print(df.loc[df["Rating"].idxmin()])

print("\nMovies with Rating Above 8")

print(df[df["Rating"] > 8])

print("\nMovies Sorted by Rating")

sorted_movies = df.sort_values(by="Rating", ascending=False)

print(sorted_movies)

sorted_movies.to_csv("Movie_Rating_Analysis.csv", index=False)

print("\nProject Completed Successfully")
```

Expected output:-

Movie Dataset

S.no	Movie	Rating	Year
0	Avatar	7.8	2009

1	Titanic	7.9	1997
2	Inception	8.8	2010
3	Interstellar	8.7	2014
4	The Dark Knight	9.0	2008

Average Rating

8.44

Highest Rated Movie:-

Movie The Dark Knight

Rating 9.0

Year 2008

Lowest Rated Movie:-

Movie Avatar

Rating 7.8

Year 2009

Movies with Rating Above 8

S.no	Movie	Rating	Year
2	Inception	8.8	2010
3	Interstellar	8.7	2014
4	The Dark Knight	9.0	2008

Project Completed Successfully.

Conclusion:-

This project demonstrates how to analyze movie ratings using Python and the Pandas library. It calculates the average rating, identifies the highest- and lowest-rated movies, filters highly rated movies, sorts the dataset, and saves the analyzed results as a CSV file.